

COSC 594

DM for Online RL

Dongsheng Ding

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Outline

- RECAP
- ONLINE POLICY OPTIMIZATION
- ONLINE DIFFUSION POLICY OPTIMIZATION

Outline

■ RECAP

Policy optimization

Policy: $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

Value function: $V^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

$P(\cdot \mid s, a) : S \times A \rightarrow \Delta_A$ – transition kernel

$s_0, a_0, r_0, s_1, a_1, \dots$ – a trajectory τ generated by a policy π

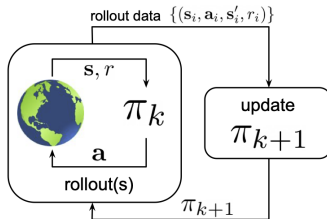
■ NONCONVEX OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho)$$

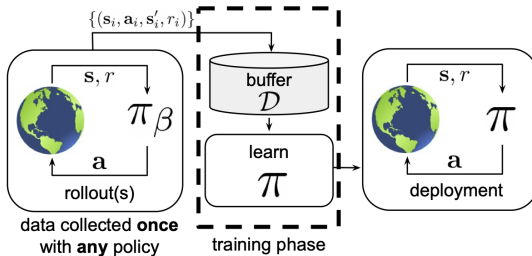
★ Search the decision in [a function space](#)

Online vs offline RL

■ ONLINE RL



■ OFFLINE RL



Actor-critic learning

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^{\pi_\theta}(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi_\theta(\cdot | s)} [Q^{\pi_\theta}(s, a)]$

■ POLICY EVALUATION

$$\phi^+ \in \operatorname{argmin}_{\phi} \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(r(s, a) + \gamma Q^{\bar{\phi}}(s', a') - \gamma Q^{\phi}(s, a) \right)^2 \right]$$

$Q^{\bar{\phi}}$ – target network w/ parameter $\bar{\phi}$

$a' \sim \pi_\theta(\cdot | s')$ – next action

■ POLICY IMPROVEMENT

$$\theta^+ \in \operatorname{argmax}_{\theta} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)] - \underbrace{\frac{1}{\eta} \mathbb{E}_{t, \epsilon_0, (s, a^0) \sim \mathcal{D}} [\|\epsilon_0 - \epsilon_\theta(a^t, t, s)\|^2]}_{\text{Behavior cloning}}$$

$$\theta \leftarrow \eta \theta + (1 - \eta) \theta^+, \phi \leftarrow \eta \phi + (1 - \eta) \phi^+$$

Policy optimization perspective in distribution space

Offline data: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β

Value function: $V^\pi(\rho) = \mathbb{E}_{s \sim \rho, a \sim \pi(\cdot | s)} [Q^\pi(s, a)]$

■ KL-REGULARIZED POLICY IMPROVEMENT

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s \sim \mathbf{d}_\rho^{\pi_\beta}, a \sim \pi(\cdot | s)} [Q^{\pi_\beta}(s, a)] - \underbrace{\eta \mathbb{E}_{s \sim \mathbf{d}_\rho^{\pi_\beta}} [D_{\text{KL}}(\pi(\cdot | s) \| \pi_\beta(\cdot | s))]}_{\text{Behavior cloning}}$$

$\Downarrow \quad \pi \leftarrow \pi_\theta, \mathbf{d}_\rho^{\pi_\beta} \leftarrow \mathcal{D}, Q^{\pi_\beta} \leftarrow Q^{\phi^+}, \text{ point-wise KL} \leftarrow \text{path-wise KL}$

$$\theta^+ \in \underset{\theta}{\text{argmax}} \mathbb{E}_{s \sim \mathbf{D}, a \sim \pi_\theta(\cdot | s)} [Q^{\phi^+}(s, a)] - \underbrace{\eta \mathbb{E}_{t, \epsilon_0, (s, a^0) \sim \mathbf{D}} [\|\epsilon_0 - \epsilon_\theta(a^t, t, s)\|^2]}_{\text{Behavior cloning}}$$

Outline

- RECAP
- ONLINE POLICY OPTIMIZATION

Policy mirror descent

$$\pi^+(\cdot | s) \leftarrow \operatorname{argmax}_{\pi'} \langle Q^\pi(s, \cdot), \pi'(\cdot | s) - \pi(\cdot | s) \rangle - \frac{1}{\eta} D_{\text{KL}}(\pi'(\cdot | s) \| \pi(\cdot | s))$$

- ★ gradient ascent over a function space
- ★ multiplicative weights update (MWU)

$$\pi^+(a | s) = \pi(a | s) \frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)}$$

- ★ fast global convergence

■ KL MINIMIZATION

$$\operatorname{minimize}_{\pi'} D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s))$$

Variational policy optimization

MWU: $\pi^+(a | s) = \pi(a | s) \frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)}$

KL minimization: $\underset{\pi'}{\text{minimize}} D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s))$

■ WEIGHTED KL MINIMIZATION

$$\begin{aligned} \underset{\pi'}{\text{minimize}} \quad & D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s)) \\ & \Updownarrow \\ \underset{\pi'}{\text{minimize}} \quad & \mathbb{E}_{a \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a)) \log \frac{\pi(a | s)}{\pi'(a | s)} \right] \end{aligned}$$

Exponential Q -weighted likelihood maximization

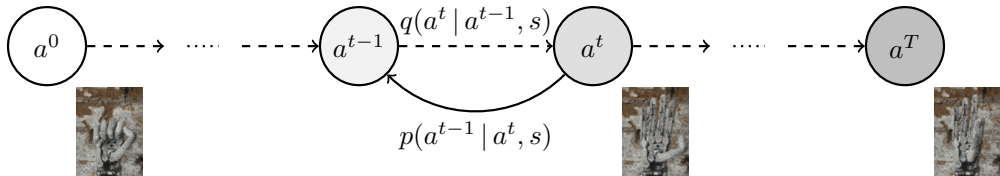
■ KL REDUCTION

$$\begin{aligned}D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s)) &= \mathbb{E}_{a \sim \pi^+(\cdot | s)} \left[\log \frac{\pi^+(a | s)}{\pi'(a | s)} \right] \\&= \mathbb{E}_{a \sim \pi(\cdot | s)} \left[\frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)} \log \frac{\pi^+(a | s)}{\pi'(a | s)} \right] \\&= \mathbb{E}_{a \sim \pi(\cdot | s)} \left[\frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)} \log \frac{\pi(a | s)}{\pi'(a | s)} \right] \\&\quad + \underbrace{\mathbb{E}_{a \sim \pi(\cdot | s)} \left[\frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)} \log \frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)} \right]}_{\pi'\text{-independent}}\end{aligned}$$

Outline

- RECAP
- ONLINE POLICY OPTIMIZATION
- ONLINE DIFFUSION POLICY OPTIMIZATION

Diffusion policy



data point a^0 – action

■ ANALOGUE W/ VAE

- ★ noiser (encoder)

$$q(a^{1:T} | a^0, s) = q(a^1 | a^0, s) q(a^2 | a^1, s) \dots q(a^T | a^{T-1}, s)$$

- ★ denoiser (decoder)

$$p(a^0 | a^{1:T}, s) = p(a^T, s) p(a^{T-1} | a^T, s) \dots p(a^0 | a^1, s) \implies \pi(a^{0:T} | s) \text{ is a RL policy!}$$

Variational policy optimization as fine-tuning

MWU: $\pi^+(a | s) = \pi(a | s) \frac{\exp(\eta Q^\pi(s, a))}{Z^\pi(s)}$

KL minimization: minimize $D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s))$

Weighted KL minimization: minimize $\mathbb{E}_{a \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a)) \log \frac{\pi(a | s)}{\pi'(a | s)} \right]$

■ POINT WEIGHTED KL MINIMIZATION

$$\text{minimize}_{\pi'} \mathbb{E}_{a^0 \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi(a^0 | s)}{\pi'(a^0 | s)} \right]$$

■ PATH WEIGHTED KL MINIMIZATION

$$\text{minimize}_{\pi'} \mathbb{E}_{a^{0:T} \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi(a^{0:T} | s)}{\pi'(a^{0:T} | s)} \right]$$

Path weighted KL

■ PATH DECOMPOSITION

$$\begin{aligned} & \mathbb{E}_{a^{0:T} \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi(a^{0:T} | s)}{\pi'(a^{0:T} | s)} \right] \\ &= \mathbb{E}_{a^{t-1} \sim \pi_t(\cdot | a^t, s), t=1, \dots, T, T+1} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi(a^{0:T} | s)}{\pi'(a^{0:T} | s)} \right] \\ &= \mathbb{E}_{a^{t-1} \sim \pi_t(\cdot | a^t, s), t=1, \dots, T, T+1} \left[\exp(\eta Q^\pi(s, a^0)) \log \prod_{t=T+1}^1 \frac{\pi_t(a^{t-1} | a^t, s)}{\pi'_t(a^{t-1} | a^t, s)} \right] \\ &= \mathbb{E}_{a^0 \sim \pi_1(\cdot | a^1, s)} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi_1(a^0 | a^1, s)}{\pi'_1(a^0 | a^1, s)} \right] \\ &\quad + \mathbb{E}_{a^{1:T} \sim \pi(\cdot | s)} \left[\sum_{t=T+1}^2 \exp(\eta Q^\pi(s, a^0)) \mathbb{E}_{a^{t-1} \sim \pi_t(\cdot | a^t, s)} \left[\log \frac{\pi_t(a^{t-1} | a^t, s)}{\pi'_t(a^{t-1} | a^t, s)} \right] \right] \end{aligned}$$

How to evaluate two weighted KL terms?

■ PREVIOUS DIFFUSION POLICY

$$\pi_t(a^{t-1} | a^t, s) = \mathcal{N}(a^{t-1}; \mu(a^t, s), \sigma^2(t)I)$$

$$\mu(a^t, a^0, s) = \frac{1}{\sqrt{\alpha_t}}a^t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon(a^t, t, s) - \text{noise-based mean}$$

$\sigma^2(t)$ – variance

■ CURRENT DIFFUSION POLICY

$$\pi'_\theta(a^{t-1} | a^t, s) = \mathcal{N}(a^{t-1}; \mu'_\theta(a^t, t, s), \sigma^2(t)I), \quad \mu'_\theta(a^t, t, s) = \frac{1}{\sqrt{\alpha_t}}a^t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon'_\theta(a^t, t, s)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(\pi_t(a^{t-1} | a^t, s) \| \pi'_\theta(a^{t-1} | a^t, s)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\epsilon(a^t, t, s) - \epsilon'_\theta(a^t, t, s)\|^2$$

■ WEIGHTED KL BETWEEN TWO GAUSSIANS

$$p = \mathcal{N}(\mu_p, \Sigma), \quad q = \mathcal{N}(\mu_q, \Sigma), \quad f(x) \in \mathbb{R}$$

$$\begin{aligned}\tilde{D}_{\text{KL}}(p \parallel q) &= \mathbb{E}_{x \sim p(\cdot)} [f(x) (\log p(x) - \log q(x))] \\&= \mathbb{E}_{x \sim p(\cdot)} \left[f(x) \left(-\frac{1}{2}(x - \mu_p)^\top \Sigma^{-1}(x - \mu_p) + \frac{1}{2}(x - \mu_q)^\top \Sigma^{-1}(x - \mu_q) \right) \right] \\&= -\frac{1}{2} \mathbb{E}_{x \sim p(\cdot)} \left[f(x) \left(2(\mu_q - \mu_p)^\top \Sigma^{-1}x + (\mu_p)^\top \Sigma^{-1}\mu_p - (\mu_q)^\top \Sigma^{-1}\mu_q \right) \right] \\&= -\frac{1}{2} \mathbb{E}_{x \sim p(\cdot)} \left[2f(x)(\mu_q - \mu_p)^\top \Sigma^{-1}(x - \mu_q) \right] \\&\quad + \frac{1}{2} \mathbb{E}_{x \sim p(\cdot)} \left[f(x)(\mu_q - \mu_p)^\top \Sigma^{-1}(\mu_q - \mu_p) \right] \\&= (\mu_p - \mu_q)^\top \Sigma^{-1} \mathbb{E}_{x \sim p(\cdot)} [f(x)(x - \mu_p)] \\&\quad + \frac{1}{2}(\mu_q - \mu_p)^\top \Sigma^{-1}(\mu_q - \mu_p) \mathbb{E}_{x \sim p(\cdot)} [f(x)]\end{aligned}$$

■ WEIGHTED KL #1

$$p = \pi'_1(a^0 | a^1, s), \quad q = \pi_1(a^0 | a^1, s), \quad f(a^0) = \exp(\eta Q^\pi(s, a^0))$$

$$\begin{aligned} & \mathbb{E}_{a^0 \sim \pi_1(\cdot | a^0, s)} \left[\exp(\eta Q^\pi(s, a^0)) \log \frac{\pi_1(a^0 | a^1, s)}{\pi'_1(a^0 | a^1, s)} \right] \\ & \approx (\epsilon_\theta(a^1, 1, s) - \epsilon'_\theta(a^1, 1, s)) \mathbb{E}_{a^0 \sim \pi_1(\cdot | a^0, s)} \left[\exp(\eta Q^\pi(s, a^0)) (a^0 - \epsilon'_\theta(a^1, 1, s)) \right] \\ & \quad + \frac{1}{2} \|\epsilon_\theta(a^1, 1, s) - \epsilon'_\theta(a^1, 1, s)\|^2 \mathbb{E}_{a^0 \sim \pi_1(\cdot | a^0, s)} \left[\exp(\eta Q^\pi(s, a^0)) \right] \end{aligned}$$

■ WEIGHTED KL #2

$$\begin{aligned} & \mathbb{E}_{a^{1:T} \sim \pi(\cdot | s)} \left[\sum_{t=T+1}^2 \exp(\eta Q^\pi(s, a^0)) \mathbb{E}_{a^{t-1} \sim \pi_t(\cdot | a^t, s)} \left[\log \frac{\pi_t(a^{t-1} | a^t, s)}{\pi'_t(a^{t-1} | a^t, s)} \right] \right] \\ & \approx \mathbb{E}_{a^{1:T} \sim \pi(\cdot | s)} \left[\sum_{t=T+1}^2 \exp(\eta Q^\pi(s, a^0)) \|\epsilon_\theta(a^t, t, s) - \epsilon'_\theta(a^t, t, s)\|^2 \right] \end{aligned}$$

Actor-critic learning

Replay buffer: $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from history

Weighted KL minimization: $\underset{\pi'}{\text{minimize}} \mathbb{E}_{a \sim \pi(\cdot | s)} \left[\exp(\eta Q^\pi(s, a)) \log \frac{\pi(a | s)}{\pi'(a | s)} \right]$

■ POLICY EVALUATION

$$\phi^+ \in \underset{\phi}{\text{argmin}} \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(r(s, a) + \gamma Q^{\bar{\phi}}(s', a') - \gamma Q^\phi(s, a) \right)^2 \right]$$

$Q^{\bar{\phi}}$ – target network w/ parameter $\bar{\phi}$

$a' \sim \pi'_\theta(\cdot | s')$ – next action

■ POLICY IMPROVEMENT

$$\theta^+ \in \underset{\theta}{\text{argmax}} \mathbb{E}_{a \sim \pi_\theta(\cdot | s)} \left[\exp \left(\eta Q^{\phi^+}(s, a) \right) \log \frac{\pi_\theta(a | s)}{\pi'_\theta(a | s)} \right]$$

$$\theta \leftarrow \eta \theta + (1 - \eta) \theta^+, \phi \leftarrow \eta \phi + (1 - \eta) \phi^+$$

Summary

■ RECAP

■ ONLINE POLICY OPTIMIZATION

Policy mirror descent, variational policy optimization

■ ONLINE DIFFUSION POLICY OPTIMIZATION

Path weighted KL minimization, actor-critic learning